Case 15: Constant rate of spread

1 Verification purpose

This isolated-process case verifies that ELMFIRE transports a circular level-set front at a prescribed, spatially uniform normal rate. It exercises raster ingestion, the spread-rate adjustment factor, the level-set update, and level-set field/spread rate output without wind, slope, crown fire, spotting, or suppression. It can reveal rate-unit errors, incorrect signs, and failure to couple the calculated rate to front motion. This is numerical verification against a known solution, not comparison with a physical fire. The verification question is whether the final spread-rate and radius satisfy the two stated tolerances at $t = 360$ s.

2 Mathematical formulation and numerical implementation

ELMFIRE advances the signed level set by

$$\frac{\partial \phi}{\partial t} + V_n |\nabla \phi| = 0,$$

with $\phi < 0$ burned and $\phi = 0$ the front. Uniform rasters and disabled directional forcing make $V_n = R_0 = 0.0116332$ m s^{-1}. The configured metric spread-rate output option makes the spread rate output use m min^{-1}, while integration time remains seconds. Raster georeferencing supplies the mesh; no obsolete computational-domain namelist is used.

3 Derivation of expected behavior

For initial circle radius $r_0 = 5$ m and constant outward speed R_0, radial symmetry reduces the level-set equation to $dr/dt = R_0$. Hence

$$r(360) = 5 + (0.0116332)(360) = 9.187952 \text{ m}, \qquad R_0 = 0.697992 \text{ m min}^{-1}.$$

These are independent consequences of the prescribed input rate and elapsed time.

The numerical reference values and the reported metrics follow directly from the generated rate raster. The raster value is $R_s = 0.0116332$ m s^{-1}, so the report-unit spread rate is

$$R_{\text{ref}} = 60 R_s = 60(0.0116332) = 0.697992 \text{ m min}^{-1}.$$

With $r_0 = 5$ m and $T = 360$ s, the exact final radius is

$$r_{\text{ref}} = r_0 + R_s T = 5 + (0.0116332)(360) = 9.187952 \text{ m}.$$

The two calculated errors are $e_R = |R_{\text{max}} - 0.697992|/0.697992$ and $e_r = |\sqrt{N_b (1 \text{ m})^2/\pi} - 9.187952|$, where N_b is the number of final cells with $\phi < 0$. Thus the acceptance substitutions are $e_R \leq 0.05$ and $e_r \leq 2$ m.

4 Verification metrics and acceptance criteria

The maximum finite final spread rate value is compared with 0.697992 m min^{-1}; a 5 percent relative tolerance catches conversion or propagation-rate errors while allowing front reconstruction effects. The

equivalent burned radius $r_A = \sqrt{N_b \Delta x^2 / \pi}$, where N_b counts valid final cells with $\phi < 0$, is compared with 9.187952 m. Area-equivalent radius is less sensitive to pixel stair-stepping than a single axial intercept; its tolerance is two cells. Nodata cells are excluded. The lexicographically last matching final level-set field and spread rate rasters are required. The case passes iff both components pass; missing or malformed required output is NOT EVALUABLE.

5 Simulation configuration and predicted outcome

The generated EPSG:32610 domain is 200 m square with 200 cells per side ($\Delta x = 1$ m), uniform fuel/moisture/terrain/weather fields, and a centered 5 m ignition circle. Duration is 360 s. The predicted final field is circular, with rate 0.697992 m min^{-1} and equivalent radius 9.187952 m.

Parameter	Configured value	Units	Role
Domain/grid	200 by 200 cells; 1	m/cell	Resolves the manufactured circle
Initial front	centered radius 5	m	Exact initial condition
Rate	0.0116332	m/s	Constant manufactured normal speed
Time control	step 1; stop 360	s	Exact final time and timestep multiple
Disabled forcing	wind, slope, crown, spotting, suppression	—	Isolates level-set transport

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

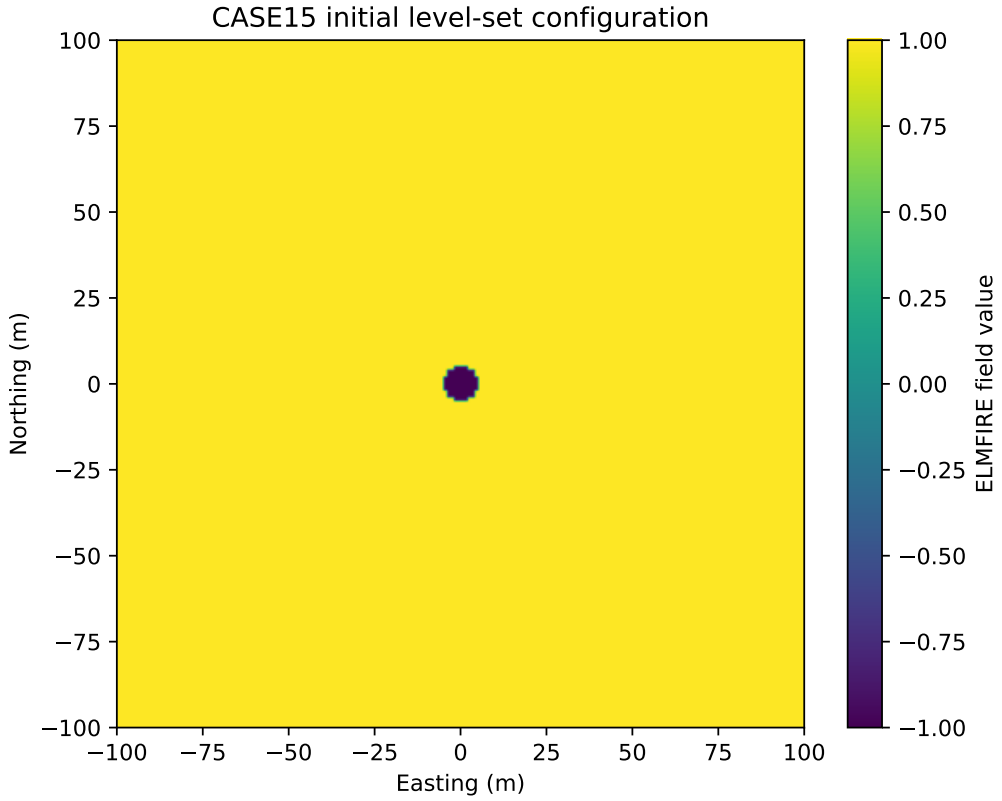


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

Only case-specific ELMFIRE rasters are read. Current status is **PASS**; 2 of 2 components pass.

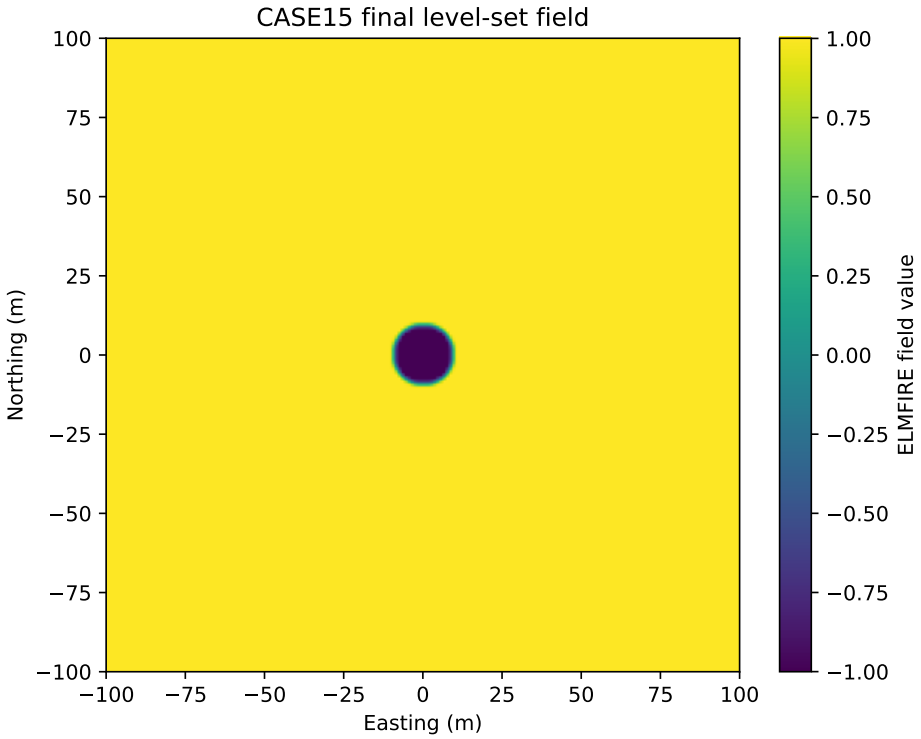


Figure 2: Final whole-domain signed level-set field for the completed simulation condition at the final output time, selected from the results of that simulation condition. Values are dimensionless signed-level-set values; the generated table supplies the acceptance limits.

7 Calculated metrics and verification decision

Metric	Expected value or limit	Calculated value	Status
maximum spread rate	0.697992 (5% relative)	0.698532 m/min	PASS
equivalent final radius	9.18795 (2 cells (2 m))	9.37302 m	PASS

The authoritative overall decision is **PASS**. The generated table is sourced directly from the case-specific calculated metrics and is not recomputed in this report.

8 Technical interpretation

A spread-rate failure points to input scaling, units, or spread rate selection. A radius-only failure indicates level-set integration or front/area discretization. A missing raster is an evidence failure, not a scientific pass.

9 Scope and limitations

This case verifies one deterministic constant-rate, flat, no-wind propagation state. It does not validate a fuel model, quantify anisotropy, or test coupled crown, spotting, or suppression behavior.